

Leyan Wu

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Education

Peking University, B.S. Candidate

Sep. 2024 – Present

- Yuanpei College, Major in Physics, Double Major in Economics.
- Core Coursework:** Data Structures and Algorithms (95+), Introduction to Computing (95+), Introduction to Computer Vision and Deep Learning (CS231n), LLM from Scratch (CS336), Fundamentals of Theoretical Physics (95+), Statistical Mechanics (95+), Methods of Mathematical Physics (95+), Modern Physics (Full Marks).
- GPA & Ranking:** 3.914 / 4.00 (Major Rank: 1/28, Top 1%)

Research Experience & Projects

Institute of Condensed Matter Physics, Peking University

Aug. 2025 – Present

- Research Topic:** Computation and Experiment on Transport Phenomena of 2D Material Heterojunctions (Advisor: Prof. Yu Ye)
- Project Highlights:** Conducted First-Principles Calculations (DFT) using **VASP** to investigate band alignment at heterojunction interfaces. Utilized **LLM Agent Tools (Codex)** to build an automated end-to-end computational workflow for parameter tuning and band structure plotting, significantly improving computational efficiency.

Basic Physics Experimental Teaching Center, PKU

Feb. 2025 – Jul. 2025

- Research Topic:** Study on Rabi Splitting and PT-Symmetry Breaking Based on Virtual Instrument Technology (Advisor: Prof. Zhi Li)
- Project Highlights:** Mastered automated experimental data acquisition. Applied **Python and time series analysis algorithms** to clean, denoise, and fit highly noisy sensor signals. Combined with simulations, successfully observed and verified the critical phase transition phenomenon of PT-symmetry breaking.

Honors, Awards & Student Leadership

- Yanchuang Group Scholarship, Peking University
- Second-Class Freshman Scholarship, Peking University
- Merit Student, Peking University
- Director of Academic & Innovation Dept.**, Peking University Student Union

Technical Skills

- AI & Programming:** Proficient in **Python (PyTorch, NumPy, Pandas)**; Familiar with C.
- Scientific Computing & Tools:** Proficient in **VASP (DFT)** and **Pymatgen** (materials informatics).
- Engineering & Deployment:** Familiar with **Linux/Shell Scripting** and **Git** version control.
- Math & Physics Foundations:** Solid foundation in theoretical physics, calculus, linear algebra, ordinary/partial differential equation solving, and numerical methods.